PRAVEEN KUMAR

Ph.D. Scholar
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Research Summary

Ph.D. researcher in autonomous mobile robotics with expertise in intelligent navigation, computer vision, semantic segmentation, robot perception, SLAM, and motion planning. Research focuses on developing robust navigation systems for autonomous ground vehicles operating in dynamic and unstructured environments. Experienced in ROS, Python, C++, Linux, LiDAR processing, RGB-D sensing, deep learning, and real-time robotic system integration.

Research Interests

- Autonomous Mobile Robots
- Intelligent Navigation
- Computer Vision
- Motion Planning
- Semantic Segmentation
- Human-Robot Interaction
- Autonomous Ground Vehicles
- Deep Learning for Robotics

Education

Indian Institute of Technology Kanpur — Kanpur, India
Ph.D. in Electrical Engineering — 2020–Present
Research Area: Intelligent Autonomous Navigation and Motion Planning for Mobile Robots
Advisors: Dr. Tushar Sandhan and Dr. K. R. Guruprasad

National Institute of Technology Rourkela — Rourkela, India
M.Tech. in Electronics and Instrumentation Engineering — 2014–2016
CGPA: 8.76

Darbhanga College of Engineering — Darbhanga, India
B.Tech. in Electrical and Electronics Engineering — 2009–2013
CGPA: 8.42

Publications

Journal Articles

1. Praveen Kumar and Tushar Sandhan, "Vision-Aided Online A* Path Planning for Efficient and Safe Navigation of Service Robots," *International Journal of Intelligent Robotics and Applications*, Springer Nature, 2026.

2. Praveen Kumar, Pratham Gupta, and Tushar Sandhan, "NeuralPathLite: Fast and Robust Diffusion-Based Path Planning for Autonomous Navigation," *IEEE Transactions on Automation Science and Engineering*, Under Revision, 2026.

Conference Publications

1. Praveen Kumar, K.R. Guruprasad, and Tushar Sandhan, "Multi-Task Visual Perception Network with LLM Conditioning for Autonomous Navigation," *IEEE World Congress on Computational Intelligence (WCCI)*, Maastricht, Netherlands, 2026.

2. Praveen Kumar and Tushar Sandhan, "Path-Tracking Control of the 4WIS4WID Electric Vehicle using Artificial Neural Networks," *IEEE ICCCNT*, 2023.

3. Praveen Kumar and U. C. Pati, "IoT-Based Monitoring and Control of Appliances for Smart Home," *IEEE International Conference on Recent Trends in Electronics, Information and Communication Technology (RTEICT)*, 2016.

4. Praveen Kumar and Umesh Chandra Pati, "Arduino and Raspberry Pi-Based Smart Communication and Control of Home Appliance System," *IEEE IC-GET*, 2016.

5. G. Sharma, Praveen Kumar, and L. Shrivastava, "An Efficient Performance of Enhanced Bellman-Ford Algorithm in Wireless Sensor Network Using K-Medoid Clustering," *Springer ICSISCET*, 2020.

6. G. Sharma, Praveen Kumar Singh, and L. Shrivastava, "Autonomous Vehicle Power Scavenging Analysis for Vehicular Ad Hoc Network," *International Conference on Intelligent Computing and Smart Communication (ICICSC)*, Springer, 2020.

Research Experience

Senior Student Research Associate Mar 2026 – Aug 2026
Indian Institute of Technology Kanpur
Project: *Simultaneous Exploration and Coverage Using a Mobile Robot*
Principal Investigator: Dr. Tushar Sandhan and Dr. K.R. Guruprasad

- Developed algorithms for simultaneous exploration and mapping.
- Evaluated autonomous navigation performance in outdoor environments.
- Implemented robotic path allocation and mission planning modules.

Student Research Associate Nov 2023 – Dec 2023
Indian Institute of Technology Kanpur
Project: *Complex Defense Object Recognition and Autonomous Handling*
Principal Investigator: Dr. Nishchal K. Verma

- Integrated LiDAR, V-SLAM and ROS based robotic perception systems.
- Developed sensor fusion and autonomous object recognition pipelines.
- Conducted outdoor robotic system testing.

Teaching Experience

Assistant Professor 2018–2020
TEQIP-III, Ministry of Education, Government of India

- Served as Assistant Professor in the Departments of Electronics and Communication Engineering.
- Taught undergraduate courses in Electronics Engineering and guided laboratory sessions.
- Mentored student projects and participated in academic and administrative activities.

Lecturer 2016–2018
College of Engineering and Technology (CET), Bhubaneswar, India

- Delivered lectures and laboratory courses in Instrumentation and Electronics Engineering.
- Guided undergraduate students in academic projects and practical training.
- Assisted in curriculum delivery, assessment, and examination activities.

Course Tutor 2021–Present
Indian Institute of Technology Kanpur

- **Introduction to Python and Agentic AI (IS901E)**
  Post Graduate Program (PGP), IIT Kanpur.
- **Introduction to Electronics (ESC201)**
  Department of Electrical Engineering, IIT Kanpur.
- **Introduction to Electrical Engineering (ESO203)**
  Department of Electrical Engineering, IIT Kanpur.

Technical Skills

Programming: Python, C++, MATLAB, C

Robotics: ROS, Gazebo, RViz, SLAM, Path Planning

Computer Vision: OpenCV, Deep Learning, Semantic Segmentation

Platforms: Linux, Ubuntu, Git, Docker

Awards and Honors

- IEEE Computational Intelligence Society Travel Grant ($1200 USD), WCCI 2026.
- M.Tech Best Project Award Nomination, NIT Rourkela.
- Qualified GATE Examination in EE and IN disciplines across multiple years.

Professional References

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